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Mark Apinis

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Boston, MA

Professional Experience

Bioinformatics Engineer I; Dana-Farber Cancer Institute, Boston, MA **Jan. 2025 - Present**

- Designed and built a full-stack web app for agentic cohort exploration and IRB submission drafting, rolling out to researchers across the institute; built on the **Claude Agent SDK** with a **FastAPI** backend, **MongoDB** database, and **React** frontend.
- Directed the design and execution of **Google Cloud** workflows to generate a multi-terabyte, AI-ready dataset of pathology features and artifact/cell segmentations from 20,000 stained WSIs, and owned development of a Jupyter notebook for expert review.
- Oversee feature and content updates for the Dana-Farber Data Catalog (DFDC), coordinating monthly data releases, improving the ETL pipeline, and developing its modern CD pipeline, with **JavaScript** and **GitHub Actions**.
- Provide technical leadership for REDCap QC and data delivery pipelines in **Python** through code reviews, participation in design discussions, and contributions such as automated data delivery that saved hundreds of hours per study.
- Manage execution of preprocessing jobs for Cancer AI Alliance projects on DFCI **OMOP** data—validating partner-provided scripts (**Python, SQL, R**) on **Databricks** and **Snowflake**, communicating with collaborators, and handling cohorts in **OHDSI ATLAS**.
- Represented DFCI at NCI's 3rd Annual Data Sharing Symposium by demoing DFDC and engaging with community leaders.

Applied Technologies Co-op, Automation Engineering; Moderna, Inc., Norwood, MA **Jan.-Jun. 2024**

- Developed scripts and integrated tools & software to automate engineering lab tasks (electrical design, 3D printing queues, label generation, Lean/5S) using **Python 3 *(pyodbc, wx)*, SQL, Git, and SmartSheet**.
- Directed the planning and development of benchtop demonstrations and supported the assembly, organization, and testing of major projects by applying control systems and mechanical engineering skills with **Productivity PLCs and SolidWorks CAD**.
- Leveraged enterprise LLMs as digital assistants to automate repetitive tasks, freeing time for problem solving.

Computational Chemistry Co-op; Novartis, AG, Cambridge, MA **Jul.-Dec. 2022**

- Built an extension enabling direct transfer of molecular data from Schrödinger Maestro to internal processing pipelines, reducing reliance on costly 3rd-party software, using **Python, pip, PowerShell, and Git**.
- Formulated algorithms and data structures to compare terabytes of chemical data for a statistical analysis study with **Python, Bash, and JupyterHub**, requiring both computer systems and molecular structure knowledge.

Software Engineering Co-op; Intuit Inc., Remote **Jun.-Dec. 2021**

- Delivered full stack production code for QuickBooks Live to facilitate reliable customer-to-expert interactions, both customer-facing with **JavaScript *(React.js, Angular, Redux, Jest)*** and administrative tools with **Java**.
- Applied engineering and teamwork strategies, such as version control with **Git**, agile development with **Jira**, integration/automation testing with **Docker and Cypress**, and real-user monitoring with **Splunk**.

Technical Skills

Programming Languages/Frameworks	**Engineering/Development Tools**	**Data and Life Science Methods**
JavaScript/TypeScript *(React.js, Node.js, Next.js)*, Python 3 *(FastAPI, Pandas, NumPy, LLMs/Agents)*, R, Bash, SQL	Agents, Google Cloud, AI-assisted development, LLM APIs, Git, SQL, MongoDB, CI/CD, Docker, testing, Agile	Machine learning/AI, cohort assembly, OMOP CDM, statistical testing, literature review and research

Education And Training

Northeastern University, Boston, MA

- **Master of Science in Bioinformatics** *(College of Science)* GPA: **3.94**/4 **August 2024**
- **Bachelor of Science in Computer Science and Biology** *(Khoury College of Computer Sciences)* **April 2023**
  Minor: **Mathematics**; Graduated *Summa Cum Laude*, GPA: **3.95**/4

Other Training: Google AI Agents Intensive *(Week-long course)*; Google AI Agents Live + Labs Boston; Google Cloud Fundamentals: Core Infrastructure; Snowflake Multimodal Document Analysis Workshop

Projects

Professor, Google Cloud: OAuth, Vertex AI, Firebase; Python 3: Agent Development Kit, Flask; TypeScript: React.js **March 2026**

- Web platform for creating college-level courses on anything a user wants, powered by AI. Agents build syllabi, generate lesson content, grade assignments, and serve as on-demand tutors. Already in use by peers to enhance studying. *(Available on request)*

MBTA Arrival Prediction Error Regression, Python, PyTorch, DuckDB, Bash **June 2026 (WIP)**

- Building a NN to predict the error in MBTA arrival predictions. Collected six weeks of prediction snapshots and associated arrivals.

Teaching Experience

Teaching Assistant; Khoury College of Computer Sciences, Northeastern University, Boston, MA

- Held office hours, proctored exams, led lab sections, wrote assignment questions, and graded.
- CS3000 Algorithms and Data (Fall 2023); CS3800 Theory of Computation (Summer 2023); CS2510 Fundamentals of Computer Science 2 (Spring 2022); CS1800 Discrete Structures (Fall 2020)